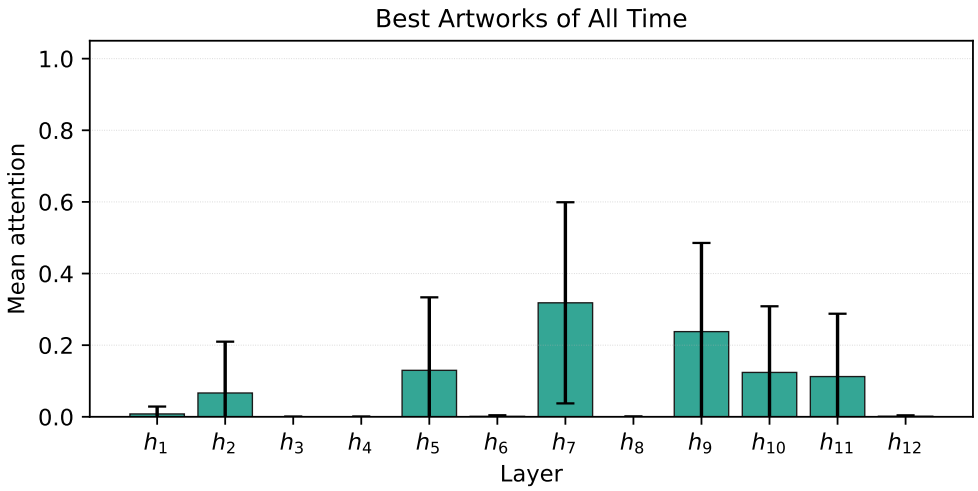

Figure 3: Layer-wise attention statistics for the *Best Artworks of All Time* dataset (mean $\pm$ standard deviation over the full test set). The attention distribution reveals heterogeneous depth contributions, with notable peaks around intermediate layers and sparse activations elsewhere.

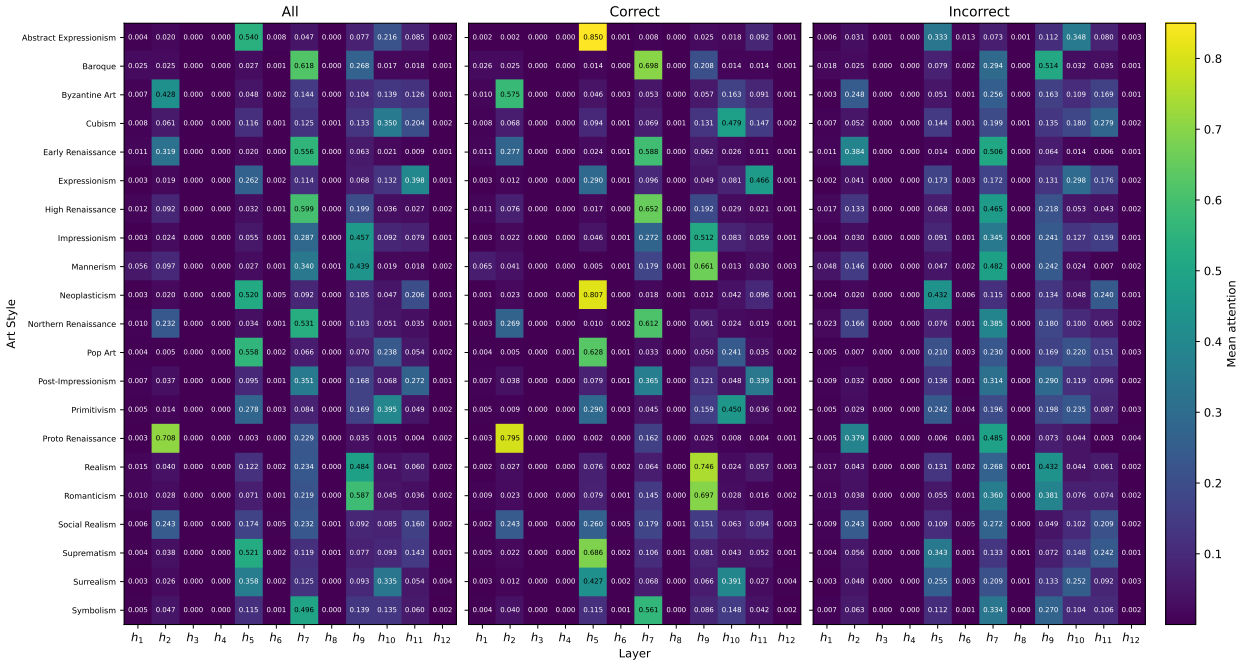

Figure 4: Class-wise mean attention profiles of LAYA on the *Best Artworks of All Time* dataset. Columns correspond to all test samples (left), correctly classified samples (center), and misclassified samples (right). Each cell reports the average attention $\bar{\alpha}_{c,i}$ assigned to layer i for class (style) c.

receive negligible attention may be redundant for the downstream task and thus candidates for simplification or removal, particularly when paired with fine-tuning or structural pruning.

Beyond compression, several directions emerge in which depth-aware output reasoning may have a broader impact. First, the observation that LAYA sometimes concentrates attention on intermediate representations provides a principled basis for *early-exit* strategies, in which inference is terminated once sufficient evidence has accumulated at earlier depths. Such adaptive computation can reduce inference costs while offering a transparent justification for the exit point. Second, systematic differences in attention profiles between correctly and misclassified samples suggest potential

diagnostic applications, such as identifying complex or out-of-distribution examples, highlighting problematic regions of the input space, or guiding dataset curation. Finally, the layer-attribution scores produced by LAYA offer new opportunities for *probing the role of depth* across architectures and tasks, providing insights into layer specialization and the degree to which different problems rely on shallow versus deep abstractions.

Overall, these findings suggest that treating the output stage as a depth-aware aggregation module—rather than a passive projection—offers a simple yet powerful way to leverage internal representations. LAYA provides a concrete demonstration of this principle and opens new opportunities in efficient inference, diagnostic interpretability, and depth-centric analysis of neural networks.

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